

SWASTIK CHOWBAY

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PERSONAL

Nationality Indian
Languages English, Italian (Basic)

RESEARCH INTERESTS

High-contrast imaging of protoplanetary disks and embedded protoplanets (VLT/SPHERE star-hopping, JWST); spectroscopic characterization of exoplanet host stars — metallicities, ages, and kinematics; Galactic chemical evolution as a driver of planet demographics; and first optical–NIR spectroscopic baselines of newly imaged accreting protoplanets.

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow Università degli Studi di Milano, Milan, Italy	July 2024 – Present
Postdoctoral Researcher Indian Institute of Astrophysics, Bengaluru, India	March 2024 – June 2024
Intern European Southern Observatory, Santiago, Chile	January 2021 – April 2021

EDUCATION

Ph.D. in Astrophysics Indian Institute of Astrophysics, Bengaluru <i>Thesis:</i> “Planet formation: From protoplanets to the main sequence” — <i>Supervisor:</i> Dr. R. K. Banyal	July 2018 – March 2024
M.Sc. in Physics University of Hyderabad, Hyderabad (CGPA: 8.67/10)	July 2016 – May 2018
B.Sc. in Physics University of Calcutta, Kolkata (70.12%)	July 2013 – May 2016

SKILLS

Languages	Python (primary), IDL, R, Java, C, C++, Fortran
Software	iSpec, IRAF, MOOG, Emcee, RADMC-3D, IRDAP, CASA (elementary)
Systems	Linux, macOS, Windows

ACHIEVEMENTS

- Qualified [NET-JRF](#) (PhD eligibility, India) 2017
- Top 15 of 11,535 in [National Graduate Physics Examination](#) 2016

PUBLICATIONS

Summary: 11 refereed publications | 6 as first author | 6 manuscripts in preparation
[Google Scholar](#): 205 citations | *h*-index: 7 | i10-index: 7 (as of May 2026)

Major research contributions

- **Stellar hosts of directly imaged planets and brown dwarfs:** showed that wide-orbit giant planets do not require metal-rich hosts, indicating a formation pathway distinct from close-in giants.
- **Galactic chemical evolution as a driver of planet demographics:** demonstrated that exoplanet host-star abundance and age trends naturally arise from Galactic chemical evolution, with high-mass-planet hosts being systematically younger.
- **Star-hopping high-contrast imaging:** delivered the first artefact-free polarimetric and total-intensity SPHERE images of the LkCa 15 disk, achieving a ~ 2 -mag contrast improvement over classical ADI/RDI.

First-author and lead-contributor publications

1. *Imaging the LkCa 15 system in polarimetry and total intensity without self-subtraction artefacts.* **C. Swastik**, Z. Wahhaj, M. Benisty, S. Arora, C. Ginski, B. B. Ren, R. G. van Holstein, R. de Rosa, R. K. Banyal, R. Tazaki. *Astronomy & Astrophysics* (2026, in press).
2. *Age analysis of exoplanet hosting stars from isochrone models.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, T. Sivarani. *Astronomical Journal* **167**, 270 (2024).
3. *Age distribution of exoplanet host-stars: chemical and kinematic age proxies from Gaia DR3.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, B. Banerjee, P. Manoj, T. Sivarani. *Astronomical Journal* **166**, 91 (2023).
4. *Galactic chemical evolution of exoplanet host-stars: are high-mass planetary systems young?* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, S. P. Rajaguru, A. Unni, B. Banerjee. *Astronomical Journal* **164**, 60 (2022).
5. *Host-star metallicity of directly imaged wide-orbit planets: implications for planet formation.* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, B. E. Reddy, S. P. Rajaguru. *Astronomical Journal* **161**, 114 (2021).
6. *Bouncing Universe with exotic radiation.* **C. Swastik**, P. K. Suresh, B. Maity. *Reports in Advances of Physical Sciences* **4**, 2050001 (2020). (M.Sc. thesis project; theoretical cosmology.)

Co-authored publications

7. *The 10 pc neighborhood of habitable zone exoplanetary systems: threat assessment from stellar encounters & supernovae.* T. Pyne, R. K. Banyal, **C. Swastik**, A. De. *Astronomical Journal* **169**, 13 (2025).
8. *Exoplanet classification through Vision Transformers with temporal image analysis.* A. Choudhary, S. Bandari, B. S. Kushvah, **C. Swastik**. *Astronomical Journal* (2025, accepted).
9. *PDS 70 unveiled by star-hopping: total intensity, polarimetry, and millimeter imaging modeled in concert.* Z. Wahhaj, M. Benisty, C. Ginski, S. Arora, **C. Swastik**, R. G. van Holstein, R. de Rosa, B. Yang, J. Bae, B. Ren. *Astronomy & Astrophysics* **687**, A257 (2024).
10. *Protoplanetary disks in K_s -band total intensity and polarized light.* B. B. Ren, M. Benisty, C. Ginski, R. Tazaki, N. L. Wallack, J. Milli, A. Garufi, J. Bae, S. Facchini, F. Ménard, P. Pinilla, **C. Swastik**, R. Teague, Z. Wahhaj. *Astronomy & Astrophysics* (2024).
11. *Carbon abundance of stars in the LAMOST-Kepler field.* A. Unni, M. Narang, T. Sivarani, M. Puravankara, R. K. Banyal, A. Surya, S. P. Rajaguru, **C. Swastik**. *Astronomical Journal* **164**, 181 (2022).

MANUSCRIPTS IN PREPARATION

As lead author

1. *Host-star metallicities and kinematics of directly imaged brown-dwarf companions.* **C. Swastik**, R. K. Banyal, M. Muduli, S. Soni, A. Jino, S. Facchini, G. Lodato, Z. Wahhaj, T. Sivarani, G. Maheswar, P. Saraf, A. Choudhary, Bhavya A. K., N. Puthiyaveetil, B. Banerjee, A. Surya, S. Biswas.
2. *First optical spectroscopy of WISPIT 2: evidence for suppressed stellar accretion.* **C. Swastik**, T. Sivarani, R. K. Banyal, S. Facchini, Z. Wahhaj, Navaneet, M. Puravankara.

3. *ERIS/NIX L'-band vortex coronagraph observations of the transitional disks J1842 and V4046 Sgr.* **C. Swastik**, S. Facchini, D. Fedele, [V. Roccatagliata](#), G. Cugno, V. Christiaens, Exo-ALMA team.
4. *A controlled comparison of confirmed exoplanet host samples from Kepler, K2, and TESS.* **C. Swastik**, S. Soni.

Co-authored

5. *Thorium abundances in FGK planet-host stars: Galactic chemical evolution and a planet-host signature.* A. Choudhary, P. Saraf, **C. Swastik**, B. S. Kushvah, R. K. Banyal, T. Sivarani, M. Gopinath.
6. *Star-hopping scattered-light imaging and combined ALMA–SPHERE modeling of the HD 163296 protoplanetary disk.* Navaneeth M. P., Z. Wahhaj, **C. Swastik**, R. K. Banyal.

SELECTED CONFERENCES AND MEETINGS

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| • Exploring Exoplanets , ICTS-TIFR, Bengaluru, India — <i>Oral</i> | Mar 2026 |
| • Spirit of Lyot 6 , Caltech, Pasadena, USA — <i>Poster</i> | Feb 2026 |
| • EXO-ELT 2025 , ESO, Garching, Germany — <i>Poster</i> | Nov 2025 |
| • GRC: Origins of Solar Systems , Mt. Holyoke, MA, USA — <i>Poster</i> | Jun 2025 |
| • GAP: Gas Accretion in Planet Formation , MPIA, Heidelberg, Germany — <i>Oral</i> | Mar 2025 |
| • Italian National Conf. on Star & Planet Formation , Sexten, Italy — <i>Oral</i> | Jan 2025 |
| • Alive Universe , Azerbaijan — <i>Oral</i> | Sep 2024 |
| • COSPAR-2024 , Busan, South Korea — <i>Oral</i> | Jul 2024 |
| • Exoplanet Conference , IISER Pune — <i>Oral</i> | Aug 2023 |
| • Towards Other Earths III , Porto, Portugal — <i>Poster</i> | Jul 2023 |
| • Protostars and Planets VII , Kyoto, Japan — <i>Poster</i> | Apr 2023 |
| • Planet–ESLAB 2023 , ESA, Netherlands — <i>Poster</i> | Mar 2023 |
| • 41st ASI Meeting , IIT Indore — <i>Oral</i> | Mar 2023 |

INVITED TALKS & RESEARCH VISITS

Each visit below included an invited departmental seminar / colloquium.

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| • TIFR , Mumbai, India — Host: P. Chaturvedi | Jan 2026 |
| • S. N. Bose National Centre , Kolkata, India — Host: R. Mondal | Dec 2025 |
| • URSC / ISRO , Bengaluru, India — Host: S. Narendranath | Sep 2025 |
| • IIA , Bengaluru, India — Host: R. K. Banyal | Aug 2025 |
| • IIT (ISM) Dhanbad , India — Host: B. S. Kushvah | Aug 2025 |
| • MIT , Cambridge, USA — Host: S. Seager | Jul 2025 |
| • NISER , Bhubaneswar, India — Host: L. Majumdar | Jul 2025 |
| • IISER Berhampur , India — Host: B. Das | Jul 2025 |
| • INAF–Arcetri , Florence, Italy — Host: D. Fedele | Feb 2025 |

AWARDS & GRANTS

- **ERC Starting Grant** — **UNVEIL** (Grant No. 101076613); postdoc funding under PI S. Facchini (UniMi). 2024–present
- **ICTS Visiting Scientist support** (ICTS/EXOPLANETS2026/03), for participation in *Exploring Exoplanets*. 2026
- **PRIN-MUR (Italy)** — **2022YP5ACE**; financial contribution. 2024
- **COSPAR Research and Travel Grant** — **~EUR 800**, for attending COSPAR-2024, Busan. 2024
- **DST International Travel Grant** — **INR 150,000**, for attending *Towards Other Earths III*, Porto. 2023
- **IIA Travel Grant** — **~EUR 3,000**, for international conference travel. 2022
- Best poster award, *Pressing for Progress: Gender Equity in Physics*. 2019

OBSERVATIONS & TELESCOPE PROGRAMS

VLT/SPHERE — awarded as PI (ESO Cycle P116, 2025–2026)

- *Revealing The Moth’s Sculptors: SPHERE’s Star-Hopping Hunt for Planets and Disk Dynamics in HD 61005* (116.29D7).
- *Unveiling Disk Substructures and Planet Formation: Scattered-Light Imaging of DoAr 33, DoAr 28, and AA Tau* (116.299M).

VLT/SPHERE — awarded as Co-PI (2024–2026)

- *Capturing dusty winds in the Orion proplyds* (116.292Y; PI: S. Facchini).
- *Rings, spirals or planets? A first clear view of the HD 141569A system* (113.269Y; PI: Z. Wahhaj).

JWST — awarded as Co-I (Cycle 5, 2026)

- *The first observation of H₂ absorption in a protoplanetary disk through transmission spectroscopy* (Program 11087; NIRSpec IFU; PI: F. M. Alarcon).

Himalayan Chandra Telescope (HCT) — 4 awarded PI programs (2020–2026)

- *Anchoring an Accreting Protoplanet: the First Optical–NIR Spectroscopic Baseline of WISPIT-2* (HCT-2026-C2-P47).
- *Long-term monitoring of the WISPIT-2 host star* (HCT 2026).
- *Determination of host-star metallicity of directly imaged planets* (HCT-2020-C2-P34).
- *Determination of host-star metallicity of directly imaged brown dwarfs* (HCT-2020-C3-P10).

Other observational work

- Awarded ~100 hours over three cycles on [HFOSC/Himalayan Chandra Telescope](#), Hanle.
- HIP 41378f transit campaign: PI for India sites ([GROWTH-India](#), Hanle; [JCBT](#), Kavalur).
- Hands-on data reduction with VLT/SPHERE (IRDAF, IRDIS), spectroscopic pipelines (IRAF/PyRAF, Astropy), and high-contrast imaging post-processing.

COMPUTING & HPC ALLOCATIONS

- **Leonardo cluster, CINECA (Italy) — 10,000 CPU-hours**; PI, long-term monitoring and modelling of the WISPIT-2 host star. 2026
- **Delphinus / Nova cluster, IIA (India) — ~10,000 CPU-hours**; PI. 2018-2024

STUDENTS MENTORED

- **Catherine John** (M.Sc., Christ University, Bengaluru) — *Demographics of exoplanets: analysis of the mass–radius relation.*
- **Aarthi Krishna** (M.Sc., IISER Tirupati) — *Understanding the brown-dwarf desert by analysing the host stars of directly imaged brown dwarfs.*
- **Athul Rathnakar** (M.Sc., St. Xavier’s University, Bangalore) — *Investigating the stellar ages of FGK stars using stellar isochrone models.*
- **Satyam Soni** (M.Sc., NIT Rourkela) — *Investigating the brown-dwarf desert for directly imaged exoplanets.*
- **Tisyagupta Pyne** (M.Sc., Visva-Bharati University, Santiniketan) — *The 10 pc stellar neighbourhood of habitable zone planetary systems.*
- **Mritunjaya Muduli** (Visvesvaraya Technological University, Belagavi) — *Computational analysis of protoplanetary disks.*
- **Anupma Choudhary** (Indian Institute of Technology (Indian School of Mines), Dhanbad) — *Thorium abundances of planet-hosting stars.*

SERVICE & OUTREACH

- **Peer review:** referee for *Astronomical Journal* (AJ) and *Astronomy & Astrophysics* (A&A).
- Student representative on IIA's outreach committee; led sky-watch programs for underprivileged children and teacher-training programs for government school teachers around Bangalore (IAU-supported).
- Co-author of thematic proposal *New Worlds: Are we alone?*, selected under [Open Astronomy Schools, IAU-100](#); volunteer with animal-welfare organizations.
- **Media coverage:** Research featured by India's [Department of Science & Technology](#) and the [Ministry of Information & Broadcasting](#).

REFEREES

- **Dr. Stefano Facchini** — Postdoc Supervisor; Assoc. Prof., UniMi; stefano.facchini@unimi.it
- **Dr. Zahed Wahhaj** — Lead Collaborator; Staff Astronomer, ESO; zwahhaj@eso.org
- **Dr. Ravinder K. Banyal** — PhD Supervisor; Assoc. Prof., IIA, Bengaluru; banyal@iiap.res.in
- **Dr. Sivarani Thirupathi** — Collaborator; Professor, IIA, Bengaluru; sivarani@iiap.res.in
- **Dr. Maheswar Gopinath** — Doctoral Committee; Assoc. Prof., IIA, Bengaluru; maheswar.g@iiap.res.in